

Programming 1: Formative Assignment Reflection

Building the Student Grade & Assignment Tracker independently was an engaging and rewarding hands-on learning experience. Translating theoretical programming concepts into a fully functioning command-line application helped solidify core Python principles like object-oriented programming, loops as well as data structures.

One of the primary technical breakthroughs was implementing Object-Oriented Programming (OOP) effectively. It was quite amazing though challenging trying to align the classes so that they co-ordinate with each other.

The project had its challenges. Initially, determining the optimal control flow for menu navigation and filter operations presented a logic hurdle. Debugging instances where input mismatches occurred led me to do some extra research in YouTube videos as well as Coursera to better understand how different functions operate.

Given additional development time, I would focus on further hardening input validation across edge cases (such as strict date string parsing and score range checks) .

Overall, this project significantly strengthened my confidence in program design, OOP principles, and self-directed problem-solving.